Macroeconomics I
Problem Set 3

1. **(Capital Utilization a la Burnside and Eichenbaum (1996)).** Consider the RBC model with variable capital utilization. The utility of the representative family is standard and given by:

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \left(\log C_t - \theta \frac{N_t^{1+\phi} - 1}{1+\phi} \right)$$

where $0 < \beta < 1$, C_t is consumption, N_t is the labor time. We will solve for the optimal equilibrium allocations.

The social planner is subject to the economy's resource constraints and respects the capital motion law. He chooses consumption, investment, household labor, and capital utilization, u_t, to maximize the utility of the representative family subject in all periods to:

$$\begin{aligned} Y_t &= C_t + I_t \\ Y_t &= Z_t (u_t K_t)^\alpha N_t^{1-\alpha} \\ K_{t+1} &= (1 - \delta_f(u_t)) K_t + I_t \end{aligned}$$

where $\delta_f(u_t)$ is a function:

$$\delta_f(u_t) = \frac{\delta}{\psi} u_t^\psi, \qquad \psi > \alpha.$$

Finally, Z_t follows a stationary AR(1) process.

(a) Solve the central planner's problem. Which conditions are *intratemporal* and which conditions are *intertemporal*?

(b) Use the FOC of u_t and show that the production function can be written as:

$$Y_t = \left(\frac{\alpha}{\delta}\right)^{\frac{\alpha}{\psi-\alpha}} Z_t^{\frac{\psi}{\psi-\alpha}} K_t^{\frac{\alpha(\psi-1)}{\psi-\alpha}} N_t^{\frac{\psi(1-\alpha)}{\psi-\alpha}}.$$

(c) Explain intuitively how the introduction of capital utilization u_t alters the amplification and persistence of the model. In particular, answer how a positive shock in Z_t changes N_t and Y_t. How does this depend on the elasticity of capital utilization ψ?[1]

[1] Hint: observe the production function above. Compare the elasticity of production with respect to the quantity of hours worked in the case with variable utilization and without variable utilization. How do wages respond to a shock Z_t?

2. **(Elastic Labor Supply in the Balanced-Growth Path)**. Consider the neoclassical growth model in discrete time. The representative household derives utility from leisure (or alternatively disutility of work), that is, in addition to the choice of consumption and capital accumulation, the household chooses the amount of hours worked h_t. The utility is:

$$\sum_{t=0}^{\infty} \beta^t \left(\log(C_t) - \theta \frac{h_t^{1+\phi}}{1+\phi} \right),$$

where $0 < \beta < 1$. Let's assume total depreciation of capital $\delta = 1$ (for simplicity). The economy's resource constraint is:

$$Y_t = K_t^{\alpha}(A_t h_t)^{1-\alpha} = K_{t+1} + C_t,$$

where technological evolution $A_{t+1}/A_t = (1+g)$. There is no population growth, $L_0 = 1$, and $A_0 = 1$. $K_0 > 0$ is given. When solving the problem, DO NOT transform the variables into efficient units (that is, work with C_t, Y_t, and K_t, the variables that are NOT stationary in the balanced-growth path).

(a) Define and solve the social planner's problem of the economy. Write down a system of three equations that together with K_0 and the TVC allow us to find the optimal allocations $\{C_t, K_{t+1}, h_t\}_{t=0}^{\infty}$ of this economy.

(b) We know that with log utility and $\delta = 1$ we can find the agent's decision rule in closed-form form. Suppose the saving rate s is constant over time: $C_t = (1-s)Y_t$. Use the method of undetermined coefficients to show that $s = \alpha\beta$ (*Hint*: use the euler equation and the resource constraint).

(c) Show that the hours worked, h_t, is a function of the parameters and does not vary over time: $h_t = h^*$.

(d) Finally, transform the variables into efficient units of labor: $k_t = K_t/A_t$, $c_t = C_t/A_t$, and $y_t = Y_t/A_t$. Find the steady-state capital, k_{ss}, as a function of the parameters and h^*. What is the growth rate of K_t in the balanced-growth path (that is, when $k_t = k_{ss}$)?

(e) For labor hours, h_t, to be constant along the balanced-growth path it is necessary for utility to have the form:

$$u(c,l) = \frac{(cv(l))^{1-\sigma} - 1}{1-\sigma},$$

where $l = 1-h$ is leisure and $v()$ a function.[2] Explain intuitively (without mathematical arguments) why h is constant in the long run, even when C_t, Y_t, and K_t have positive growth rates (*Hint*: think about what should happen with the income effect, the substitution effect, and complementarity of consumption and leisure for the amount of demanded leisure to remain constant).

3. **(Money in Utility Function).**[3] Consider the following RBC model with money in the utility function. Households value money in their utility (you can interpret this as a need for money for transactions or liquidity preference). The utility is given by:

[2] In our case, h_t is also constant outside of steady state. This is due to the assumptions of log and $\delta = 1$. In general, this does not happen.

[3] Based on Walsh (2010, chap. 2)

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \left(\log C_t - \theta \frac{N_t^{1+\phi} - 1}{1+\phi} + \frac{(M_t/P_t)^{1-\nu} - 1}{1-\nu} \right)$$

where $0 < \beta < 1$, C_t is consumption, N_t is the labor time, P_t is the price of the final good (in units of money), and M_t/P_t is the quantity of real money holdings of the household (or real balances). Households choose how much to consume, C_t, how much to work, N_t, how much money to hold, M_t, how much capital to invest K_{t+1}, and how many bonds to buy B_{t+1}:

The household's budget constraint is:

$$P_tC_t + P_t(K_{t+1} - (1-\delta)K_t) + B_{t+1} + M_t = P_tw_tN_t + P_t\hat{r}_tK_t + (1 + i_{t-1})B_t + M_{t-1} + P_tT_t$$

Where i_t is the nominal interest rate paid on bonds and T_t is a lump-sum transfer paid by the government. Government transfer is financed via money printing:

$$P_tT_t = M_t - M_{t-1}.$$

Production is standard and follows a Cobb-Douglas function: $Y_t = K_t^{\alpha}N_t^{1-\alpha}$, where $\alpha \in (0,1)$. The household is subject to a no-Ponzi scheme constraint. Define $P_{t+1}/P_t \equiv 1 + \pi_t$ where π_t is the inflation rate of the economy.

(a) Describe and solve the household's problem. Write the demand for real balances, M_t/P_t, as a function of C_t and i_t. How does it depend on i_t and C_t?

(b) Find the Fisher equation $i_t = \mathbb{E}_t r_{t+1} + \pi_{t+1}$, where $r_t \equiv \hat{r}_t - \delta$.

(c) Suppose that in the steady state the money growth rate is $M_{t+1}/M_t = 1 + \mu$ and that real balances are constant in the steady state $M_t/P_t = M_{t+1}/P_{t+1} = m$. Find a system of (eight) equations that solves the problem in the steady state for the variables: (r, w, K, N, C, i, m, T).

(d) Suppose an increase in the money growth rate: μ. How does this affect the variables in the steady state? How do changes in M_t affect the price level and other endogenous variables? What would happen if agents expected an increase in the money supply in $t+1$?